

ASTRO RAIDER

Suit System — Game Design Document

Field	Value
Document	Suit System GDD
Project	Astro Raider
Date	23.05.2026
Engine	Godot 4 / GDScript
Purpose	Design, balancing, technical structure, and open issues of the Suit System.

1. Suit System Overview

The Suit System is the meta-progression and loadout foundation of Astro Raider. Each suit defines not just starting stats, but also long-term level bonuses, perk scaling, unlocks, and optional runtime mechanics such as suit-specific weapon logic.

Key principles:

- Every player starts a run with exactly one selected suit.
- The suit generates the player's runtime stats before movement, perks, and UI access them.
- Suit EXP is collected during a run and written to the suit only at run-end or Game Over.
- Perks can scale against final stats or explicitly against suit-scaled base values.
- Suit levels unlock weapon upgrades and future meta-content.

2. Current Suits

Two suits are currently implemented. Both are unlocked by default.

Key	Suit	Role	Unlock	Status
1	Trailblazer	Balanced exploration suit with scaling Photon Gun mechanic.	has_unlocked = true	Full
2	Bloodreaver	Aggressive combat suit with very high base stats.	has_unlocked = true	Data only — no SuitMechanics node

2.1 Trailblazer

The starter suit. Balanced across all stats with a unique weapon that evolves as the suit levels up. The Photon Gun unlocks additional fire modes at milestone levels, rewarding players who invest in the suit's meta-progression.

Stat	Base Value
Gravity Strength	200
Max Speed	250
Bohrer Damage	1
Bohrer Knockback	3
Max HP	100
Armor	1
Crit Chance	0
Projectile Damage	0

Stat	Base Value
Attack Cooldown	0.4s

Photon Gun — Weapon Milestones

The Trailblazer's weapon is a Photon Gun that grows with the suit level. Alt Fire unlocks at Level 20 and is gated behind suit progression.

Suit Level	Unlock
1–19	Normal Fire: 1 projectile
20	Alt Fire unlocked: charged 3-shot burst
40	Normal Fire fires 2 parallel projectiles
60	Alt Fire becomes a 5-shot burst
80	Normal Fire fires 3 parallel projectiles

Alt Fire Details

Property	Value
Input	Right mouse button
Charge Time	0.15s minimum, 1.0s maximum
Visual Feedback	Aim sprite scales from 0.5 to 0.85 during charge
Level 20 Spread	3 shots at 13°
Level 60 Spread	5 shots at 10°

2.2 Bloodreaver

An aggressive high-risk suit built around combat stats. With 500 HP, 50% crit chance, and a 0.2s attack cooldown, Bloodreaver is significantly stronger than Trailblazer out of the box. It currently has no dedicated SuitMechanics node — it contributes through raw stats and serves as the required suit for Bloodreaver-themed perks like Vampire Bite and Blood Claws.

Stat	Base Value
Gravity Strength	250
Max Speed	300
Bohrer Damage	10
Bohrer Knockback	5
Max HP	500
Armor	2
Crit Chance	50%
Projectile Damage	9
Attack Cooldown	0.2s

Note: Bloodreaver's base stats are extremely powerful. Before v1.0 it should either have higher unlock requirements or adjusted base values — especially HP 500, Crit 50, and Attack Cooldown 0.2.

3. EXP & Meta-Progression

Suit EXP is earned during a run from multiple sources and applied to the selected suit at run-end. Leveling a suit permanently improves its base stats and unlocks new weapon capabilities.

3.1 Level Formula

EXP required to reach the next level is calculated as:

```
ceil((exp_base + pow(level, exp_curve_power) * exp_linear_gain) * exp_multiplier)
// Default: ceil(75 + level^1.35 * 18)
```

Suit Level	EXP to Next Level	Total EXP from Level 1
1	93	—
10	478	2,196
20	1,103	9,668
40	2,694	46,212
60	4,602	117,774
80	6,750	229,865
100 (max)	8,975	386,860

3.2 EXP Sources

Source	Amount	Implementation
Enemy Kill	EnemyStats.suit_exp_reward per enemy	EnemyBaseTemplate → add_kill_exp()
Boss Kill	500 (default)	BossEntity → add_boss_exp()
Wave Clear	10 + wave_number × 2	EntitySpawner.give_wave_exp()
Generator Defense	5 + wave_number × 1	EntitySpawner.give_generator_defense_exp()
Mining	10 tiles destroyed = 1 EXP	environment_tiles → add_mining_tiles()
Overflow Crystals	1 surplus crystal = 1 EXP	perks_manager → add_overflow_crystals()

Enemy EXP Rewards

Enemy	Suit EXP
Voidling	2

Enemy	Suit EXP
Bat	3
Demon Eye	5
Glimblob	7
Noxul — the Void Parasite (Boss)	500

3.3 Run Flow

How EXP moves from a run into the suit:

- `main_game.gd` calls `SuitExpRunTracker.reset_run_exp()` at run start.
- All sources add to `current_run_exp` and the `exp_breakdown` dictionary during the run.
- On Game Over, `healthbar.gd` applies the collected EXP to the current player's suit.
- On run completion (`ends_run RoundType`), EXP is applied to all active player suits.
- `run_exp_applied` signal fires, then the tracker resets.

4. Stat System & Level Rules

Suit level bonuses are not written directly into base values. Instead, `SuitData` duplicates the `Stats` resource and adds a modifier with the ID `suit_<Key>`. This keeps suit bonuses, perk bonuses, and base values fully separate and non-destructive.

4.1 Available Stat Keys

StatKey	Applied to
MAX_HP	<code>max_hp_modifiers</code>
MAX_SPEED	<code>max_speed_modifiers</code>
GRAVITY_STRENGTH	<code>gravity_strength_modifiers</code>
ROTATION_SPEED	<code>rotation_speed_modifiers</code>
PROJECTILE_DAMAGE	<code>projectile_damage_modifiers</code>
ATTACK_SPEED	<code>attack_speed_modifiers</code>
CRIT_CHANCE	<code>crit_chance_modifiers</code>
BOHRER_DAMAGE	<code>bohrer_damage_modifiers</code>
INVINCIBILITY_FRAME	<code>hit_iframe_duration_modifiers</code>
PROJECTILE_LIVES	<code>projectile_lives_modifiers</code>
STUN_STRENGTH	Defined in enum — not yet matched in <code>SuitData</code>
REVEAL_RADIUS_TILES	Defined in enum — not yet matched in <code>SuitData</code>

4.2 Trailblazer Level Rules

Level bonuses fire via repeatable rules defined in the SuitData resource. Rules trigger based on a start level and a repeat interval.

Rule	Start Level	Every N Levels	Bonus per Trigger
Base Progression	2	1	+1 Max HP, +1 Max Speed
Projectile Bonus	3	3	+1 Projectile Damage
HP Bonus	5	5	+2 Max HP
Decade Milestone	10	10	+2 Projectile Damage, +1 Rotation Speed, +1 Bohrer Damage

Cumulative Bonuses at Key Levels

Suit Level	HP Bonus	Speed Bonus	Proj. Damage	Rotation Speed	Bohrer Damage
1	+0	+0	+0	+0	+0
10	+13	+9	+5	+1	+1
20	+27	+19	+10	+2	+2
40	+55	+39	+21	+4	+4
60	+83	+59	+32	+6	+6
80	+111	+79	+42	+8	+8
100	+139	+99	+53	+10	+10

5. Perk Integration

Several perks use the selected suit as a scaling base. They call `get_selected_suit_modifier_id()` and use `get_suit_scaled_*` getters so that percentage bonuses scale against suit base + suit level bonus, but not recursively against other perk bonuses.

Perk	Suit Scaling
Construction Expert	Bohrer Damage scales with suit-scaled Bohrer Damage; min bonus = perk level
Power Shot	Projectile Damage scales with suit-scaled Projectile Damage; min bonus = perk level
Extra Health	Max HP bonus = suit-scaled Max HP × perk percent
Speed It Up	Max Speed and Gravity Strength scale with suit-scaled values
Energy Overload	Bohrer Damage, Max Speed, and Attack Cooldown use suit-scaled values
Barrier Shield	Shield HP = max(fixed perk value, suit-scaled Max HP × level percent)
Each Round Heal	Heals a percent of suit-scaled Max HP at wave end

5.1 Suit-Level Meta-Gates

Perks can require a specific suit and minimum suit level before appearing in the selector. Currently the gate logic exists but is bypassed because `starts_unlocked` defaults to `true` in `Perk_Resource.gd`.

Perk	Required Suit	Current Issue
Vampire Bite	Bloodreaver	<code>starts_unlocked = true</code> by default — gate is bypassed
Blood Claws	Bloodreaver	<code>starts_unlocked = true</code> by default — gate is bypassed

Fix: Set `starts_unlocked = false` in the Perk Resource for these perks, or adjust the gate check logic in `PerkSelection`.

6. Save System

Aspect	Detail
Save Slots	GameSaver manages 5 slots. Default is Slot 1.
Files	<code>user://saves/save_slot_<slot>.cfg</code> for meta/suit progress
Encryption	<code>ConfigFile.save_encrypted_pass()</code> with <code>SAVE_PASSWORD</code>
Suit Sections	Each suit stored in section <code>suit_<key></code> with <code>current_level</code> , <code>current_exp</code> , <code>has_unlocked</code>
Migration	<code>_migrate_save_data()</code> restores missing meta keys and suit sections
Default State	Both Trailblazer and Bloodreaver start as <code>has_unlocked = true</code>
Unlock API	<code>unlock_suit(suit_key)</code> sets <code>has_unlocked</code> and saves immediately

7. Adding a New Suit

Follow these steps when creating a new suit:

- Add the new suit key to the `SuitKeys` enum in `SuitData.gd`.
- Register the resource path in `SuitData.Keys_res`.
- Create `resources/characters/Suit_<Name>.tres` with `SuitData`, `Stats`, unlock state, and optional `stat_level_rules`.
- If the suit has custom weapon or active mechanics: extend `SuitMechanics` into a new script.
- Add a node under `Player/SuitsManager` in `player.tscn` and set the `suit_key`.
- If new `StatKey` values are needed, extend `SuitData._apply_level_bonuses_to_stats()`.
- Configure perks with `required_suit` / `required_suit_level` and set `starts_unlocked = false` for the gate to work.

- Update the suit selection UI and save logic if players should be able to switch suits.

8. Known Issues & Open Points

Issue	Severity	Details
Bloodreaver has no SuitMechanics node	Medium	No suit-specific weapon logic activates when Bloodreaver is selected.
STUN_STRENGTH / REVEAL_RADIUS_TILES not applied	Low	Both StatKeys exist in the enum but are not matched in SuitData._apply_level_bonuses_to_stats().
Perk meta-gates bypassed	Medium	unlock_type = SUIT_LEVEL is ignored because starts_unlocked = true. Needs explicit false in resource.
Run EXP is global	Low	SuitExpRunTracker is shared for the whole run. Multiplayer distribution resets after first apply.
Duplicate selected_suit sources	Low	GameSaver.selected_suit and PlayerDataBuilds.selected_suit coexist. Canonical source not defined.
Suit selection UI not implemented	Medium	Players currently have no in-game screen to switch between suits.
Bloodreaver base stats very high	High	HP 500, Crit 50%, Attack Cooldown 0.2 — needs unlock gate or stat reduction before v1.0.

9. File Reference

File	Role
scripts/resources/player/SuitData.gd	Core resource: EXP formula, level-ups, runtime stats, suit loading
scripts/resources/player/SuitStatLevelRule.gd	Repeatable level rules
scripts/resources/player/SuitsStatBonus.gd	Individual stat bonuses and StatKey enum
scripts/resources/player/SuitExpRunTracker.gd	Run EXP accumulation, breakdown, apply-to-suit
scripts/managers/GameSaver.gd	Save slots, suit progress, unlocks, save signals
scripts/actors/player/player.gd	Load selected suit and apply stats
scripts/actors/player/suits/suits_manager.gd	Switch active SuitMechanics
scripts/actors/player/suits/suit_mechanics.gd	Base class for suit-specific runtime logic
scripts/actors/player/suits/trail_blazer.gd	Trailblazer: Photon Gun, Alt Fire, weapon level milestones
resources/characters/Suit_Trailblazer.tres	Trailblazer data, level rules, icon
resources/characters/Suit_Bloodreaver.tres	Bloodreaver data and base stats

File	Role
scripts/perks/resources/Perk_Resource.gd	Meta unlock by suit level
scripts/ui/level_frame.gd	Suit level display in HUD
scripts/ui/healthbar.gd	Game Over EXP apply and HP sync from suit stats